

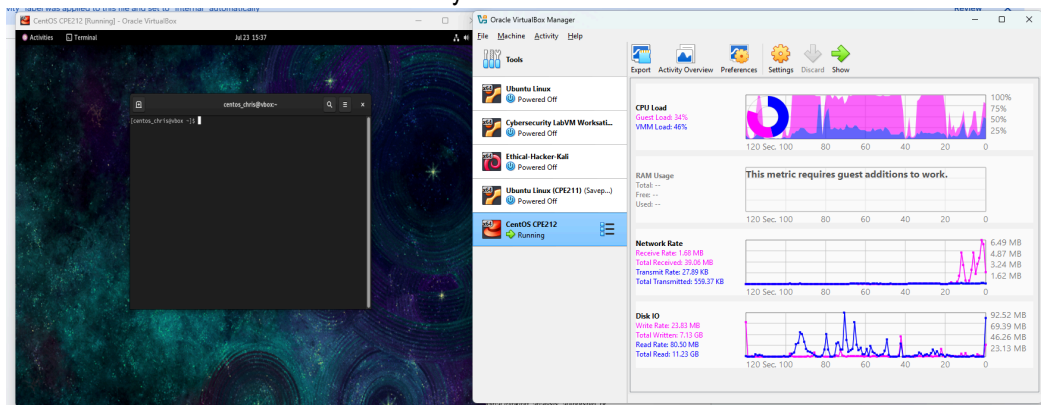
COURSE CODE/TITLE:	CPE212 - Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	07/23/2026	DATE SUBMITTED:	07/23/2026
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ACTIVITY NO. 3.1 Install SSH Server on CentOS or RHEL 8

1. PROCEDURE OUTPUT

Task 1: Download the CentOS or RHEL-8 image

1. Download the image of the CentOS here:
2. Create a VM machine with 2 Gb RAM and 20 Gb HD.
3. Install the downloaded image.
4. Show evidence that the OS was installed already.



Task 2: Install the SSH server package openssh

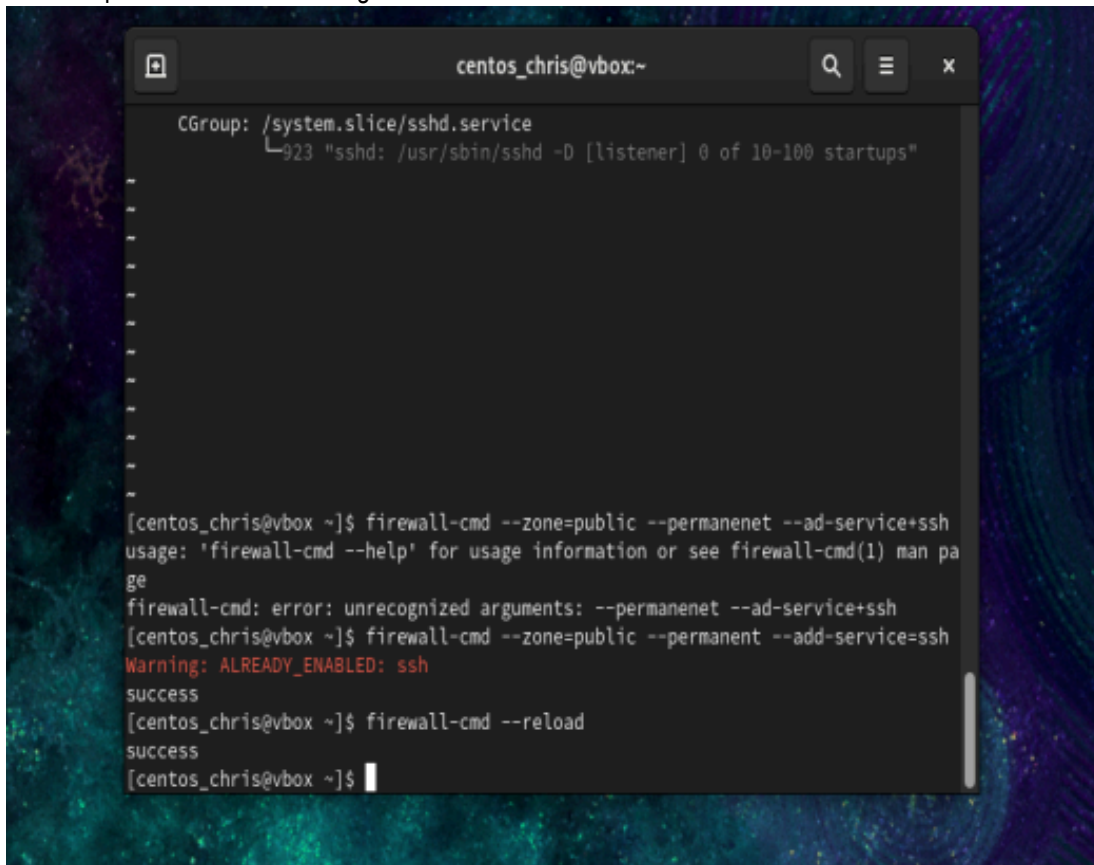
```

centos_chris@vbox:~ — systemctl status sshd
CentOS Stream 9 - BaseOS                1.7 MB/s | 9.0 MB    00:05
CentOS Stream 9 - AppStream              2.5 MB/s | 28 MB     00:11
CentOS Stream 9 - Extras packages        13 kB/s | 21 kB      00:01
Package openssh-server-9.9p1-10.el9.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[centos_chris@vbox ~]$ systemctl start ssh
Failed to start ssh.service: Unit ssh.service not found.
[centos_chris@vbox ~]$ systemctl start sshd
[centos_chris@vbox ~]$ systemctl enable sshd
cp[centos_chris@vbox ~]$ systemctl status sshd
• sshd.service - OpenSSH server daemon
   Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; preset: enab
   Active: active (running) since Thu 2026-07-23 15:36:06 PST; 9min ago
     Docs: man:sshd(8)
           man:sshd_config(5)
   Main PID: 923 (sshd)
     Tasks: 1 (limit: 10502)
    Memory: 1.7M (peak: 1.9M)
       CPU: 10ms
      CGroup: /system.slice/sshd.service
              └─923 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

lines 1-11/11 (END)

```

Open the SSH port 22 to allow incoming traffic:



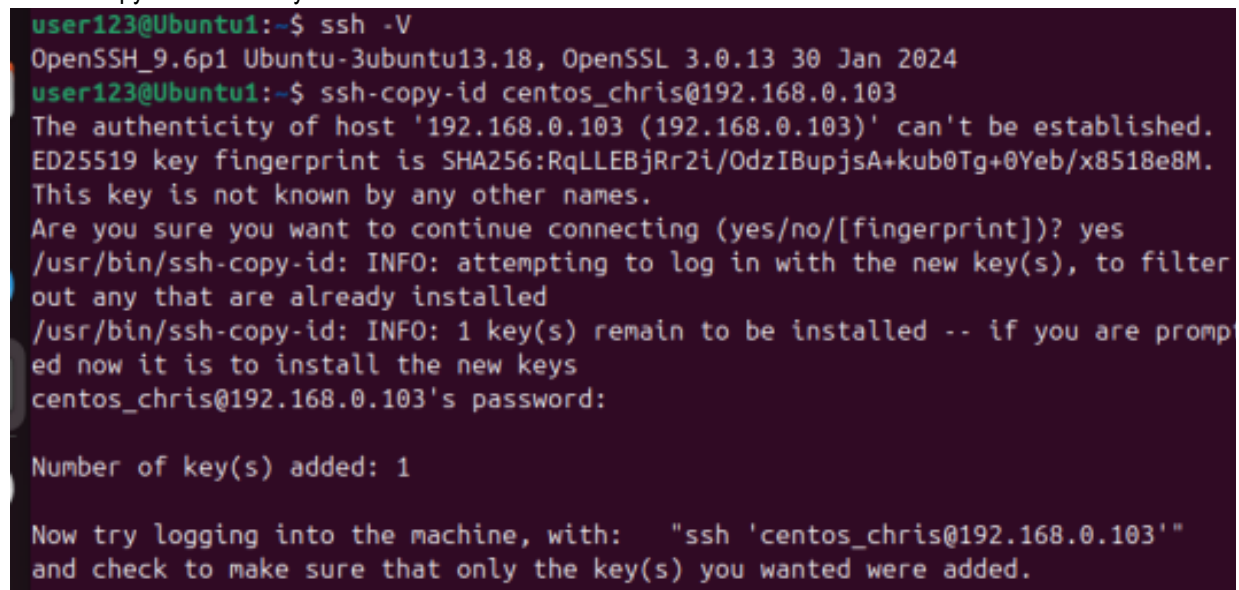
```

centos_chris@vbox:~
CGroup: /system.slice/sshd.service
└─923 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

[centos_chris@vbox ~]$ firewall-cmd --zone=public --permanenet --ad-service+ssh
usage: 'firewall-cmd --help' for usage information or see firewall-cmd(1) man page
firewall-cmd: error: unrecognized arguments: --permanenet --ad-service+ssh
[centos_chris@vbox ~]$ firewall-cmd --zone=public --permanent --add-service=ssh
Warning: ALREADY_ENABLED: ssh
success
[centos_chris@vbox ~]$ firewall-cmd --reload
success
[centos_chris@vbox ~]$

```

Task 3: Copy the Public Key to CentOS



```

user123@Ubuntu1:~$ ssh -V
OpenSSH_9.6p1 Ubuntu-3ubuntu13.18, OpenSSL 3.0.13 30 Jan 2024
user123@Ubuntu1:~$ ssh-copy-id centos_chris@192.168.0.103
The authenticity of host '192.168.0.103 (192.168.0.103)' can't be established.
ED25519 key fingerprint is SHA256:RqLLEBjRr2i/OdzIBupjsA+kub0Tg+0Yeb/x8518e8M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are promp
ed now it is to install the new keys
centos_chris@192.168.0.103's password:

Number of key(s) added: 1

Now try logging into the machine, with:  "ssh 'centos_chris@192.168.0.103'"
and check to make sure that only the key(s) you wanted were added.

```

```
centos_chris@localhost:~$ ifconfig
[centos_chris@localhost ~]$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.0.103 netmask 255.255.255.0 broadcast 192.168.0.255
    inet6 fe80::a00:27ff:fe57:4827 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:57:48:27 txqueuelen 1000 (Ethernet)
    RX packets 95 bytes 12283 (11.9 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 92 bytes 9800 (9.5 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 18 bytes 2112 (2.0 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 18 bytes 2112 (2.0 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[centos_chris@localhost ~]$ cat ~/.ssh/authorized_keys
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIN/qsrVhVs/OtIvHkDgYZDZYDmhkU/gk/ogVqdJVe/6
user123@Ubuntu1
[centos_chris@localhost ~]$
```

Task 4: Verify ssh remote connection

```
Ubuntu Linux (Savepoint1) [Running] - Oracle VirtualBox
centos_chris@localhost:~$ ssh-copy-id centos_chris@192.168.0.103
The authenticity of host '192.168.0.103 (192.168.0.103)' can't be established.
ED25519 key fingerprint is SHA256:RqLLEBjRr2i/OdzIBupjsA+kub0Tg+0Yeb/x8518e8M.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
centos_chris@192.168.0.103's password:
Number of key(s) added: 1
Now try logging into the machine, with: "ssh 'centos_chris@192.168.0.103'"
and check to make sure that only the key(s) you wanted were added.
user123@Ubuntu1:~$ ssh centos_chris@192.168.0.103
Activate the web console with: systemctl enable --now cockpit.socket

Last login: Thu Jul 23 16:15:22 2026
[centos_chris@localhost ~]$
[centos_chris@localhost ~]$ whoami
centos_chris
[centos_chris@localhost ~]$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. What do you think we should look for in choosing the best distribution between Debian and Red Hat Linux distributions?

- It should be what you need it for. Like your system requirements, user requirements, and why you would use that specific distro. Since a lot of distro are for specific/specialized use cases. So it definitely depends on your own needs.

2. What are the main differences between Debian and Red Hat Linux distributions?

- It's their package manager. With debian it uses the apt package manager and with RHEL distros it uses dnf/rpm.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.